

f00-sh

joule distributed cluster

STANDARD OPERATING PROCEDURE - joule cluster

Document: JOULE-SOP-OPS-001

Date: 2026-08-02

Version: v0.1.8

Standard: NASA NAI 1450.3 memorandum layout

Reply to Attn of: OPS/JOULE

TO: joule operators and release owners

FROM: Project Engineering

SUBJECT: joule operator standard operating procedure

REF: AGENTS.md, README.md, docs/design/cluster-v0.md

Operators run joule as a GUI-first local pool or multi-agent cluster: start control, donate with an agent under local policy, watch live capacity graphs, and ship only when cargo gates and the docs pack are green.

1. Purpose

This SOP defines day-to-day operation of a joule lab or standing pool: quality gates, donor contribution control, live capacity visibility (CLI and GUI), and the release ship flow for f00-sh/joule.

2a. Native installers

Prefer OS installers from GitHub Releases: Windows Setup.exe, macOS .pkg/.dmg (joule.app), Linux .deb. CLI curl/irm one-liners remain for automation.

- Windows: joule-*-windows-x86_64-setup.exe
- macOS: joule-*-darwin-*.pkg or .dmg
- Linux: sudo dpkg -i joule-*-linux-*.deb

2. Day-to-day operator gates

Before exposing any gateway or telling donors the pool is open, complete these checks in order. Fail closed on any red step.

2.1 Run workspace tests.

```
cargo test --workspace
```

2.2 Optional offline lab.

```
cargo run -p joule --release -- lab
```

2.3 Start control (GUI or CLI).

```
joule gui # or: joule control
```

2.4 Start agent with local policy.

```
joule agent --control 127.0.0.1:7701 --policy ~/.config/joule/donor-policy.json
```

2.5 Read live capacity.

```
joule capacity --api http://127.0.0.1:7700 --json
```

2.6 Confirm GUI Overview shows backends and connection state; Graphs zoom/pan work.

Figure 1. Day-to-day operator gates

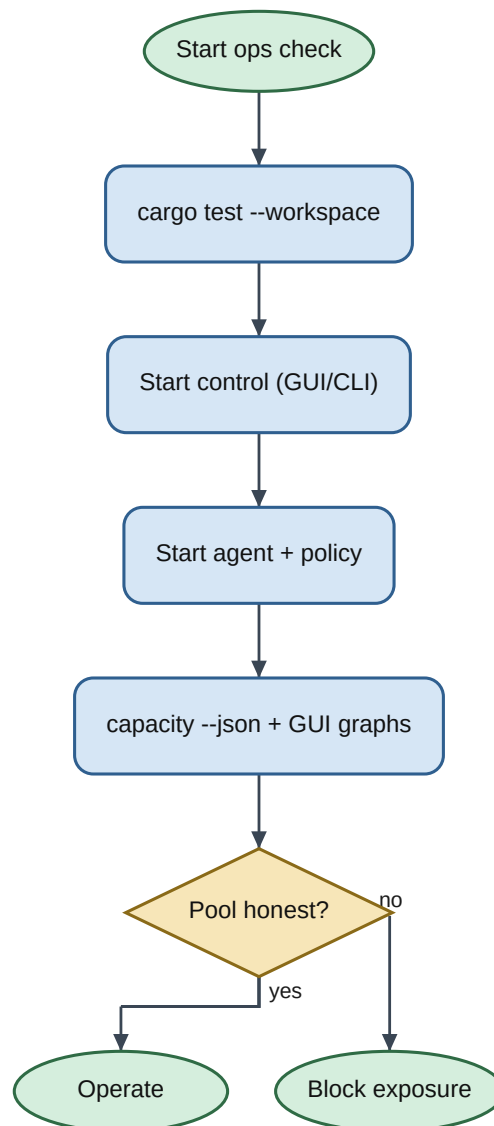


Figure 2. GUI-first operator surface



joule version and GUI tab path (Overview / Graphs / Donor / Chat) after release build.

3. Donor local controls (law 5)

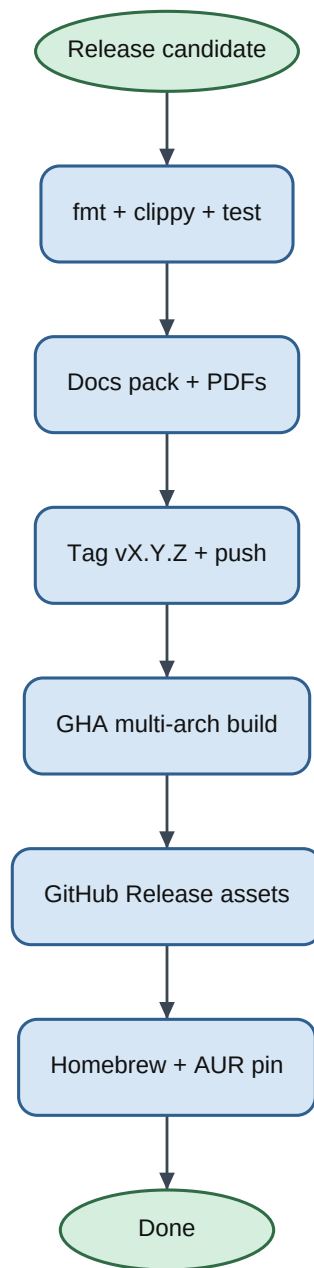
Remote operators cannot raise local caps. Agents reload the policy path passed with --policy. Schedule windows use OS local timezone (tm_gmtoff on Unix).

- Pause/resume: joule donor pause|resume or GUI Donor tab / tray
- VRAM cap: joule donor set-cap N (0 clears) or tray --donor-set-cap
- Schedule / thermal / battery: donor policy JSON or GUI Donor tab
- Status: joule donor status - prints local schedule and sensors

4. Release ship flow

Ship only when gates are green. Documentation pack is mandatory: CHANGELOG, file_id.diz, README, Pages, man, SOP PDF, this-version NASA memo PDF, SemVer tag, GitHub Release assets (CI on tag v*).

Figure 3. Release ship flow



5. Product laws (do not violate)

- Compute is the only currency on the public pool - no money
- No contribution ? no API access
- Credits = sealed hash-chain millijoules only; verified VRAM only for mint/placement
- Single model: kimi-open

- Website is not a weight CDN - content-addressed peer seed only

Project Engineering
f00-sh / joule OPS

Enclosure: file_id.diz; CHANGELOG.md; docs/releases/v0.1.7-memo.pdf
cc: Maintainers, Release Operators